SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

Annexure A

Resolution Algorithm Implementation

Report: Resolution Algorithm

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1. Rain and Wet Ground

1.1 Knowledge base and propositional representation

Let:
- R**:** "It rains"
- W**:** "The ground is wet"

KB:
1. $R \rightarrow W$
2. R

Goal: Prove W.

1.2 CNF conversion and negated goal

- From $R \rightarrow W$: $\neg R \vee W$→ **C1**
- Fact R: R→ **C2**
- Negated goal: $\neg W$→ **C3**

Clause set: C1: $\neg R \vee W$C2: RC3: $\neg W$

1.3 Resolution steps
1. **C1 & C2:** $(\neg R \vee W)$and (R)→ resolve on R: **C4:** W
2. **C4 & C3:** $(W)$and $(\neg W)$→ resolve on W: **C5:** □

1.4 Conclusion

Empty clause □ obtained ⇒ W("the ground is wet") is logically entailed by the KB using resolution.

2. Student Assignment Submission

2.1 Propositional representation

Let:

- S_R**:** "Rahul submitted the assignment"
- M_R**:** "Rahul receives internal marks"

General rule: "If a student submits the assignment, then the student receives internal marks" instantiated for Rahul:

KB:
1. $S_R \rightarrow M_R$
2. S_R

Goal: Prove M_R.

2.2 CNF and negated goal

- From $S_R \rightarrow M_R$: $\neg S_R \vee M_R$→ **C1**
- Fact S_R: S_R→ **C2**
- Negated goal: $\neg M_R$→ **C3**

Clause set: C1: $\neg S_R \vee M_R$C2: S_RC3: $\neg M_R$

2.3 Resolution steps
1. **C1 & C2:** → resolve on S_R: **C4:** M_R
2. **C4 & C3:** → resolve on M_R: **C5:** □

2.4 Conclusion
Empty clause □ ⇒ M_R("Rahul receives internal marks") is proved by resolution.

3. Library Membership

3.1 Propositional representation

Let:
- L_P**:** "Priya is a library member"
- B_P**:** "Priya can borrow books"

KB:
1. $L_P \rightarrow B_P$
2. L_P

Goal: Prove B_P.

3.2 CNF and negated goal

- From $L_P \rightarrow B_P$: $\neg L_P \lor B_P$→ **C1**
- Fact L_P: L_P→ **C2**
- Negated goal: $\neg B_P$→ **C3**

Clause set: C1: $\neg L_P \lor B_P$C2: L_PC3: $\neg B_P$

3.3 Resolution steps

1. **C1 & C2:** → resolve on L_P: **C4:** B_P
2. **C4 & C3:** → resolve on B_P: **C5:** □

3.4 Conclusion

Empty clause □ ⇒ B_P(“Priya can borrow books”) is logically entailed.

4. Placement Eligibility

4.1 Propositional representation

Let:

- A_A**:** “Arun cleared the aptitude test”
- E_A**:** “Arun is eligible for placement”

KB:

1. $A_A \rightarrow E_A$
2. A_A

Goal: Prove E_A.

4.2 CNF and negated goal

- From $A_A \rightarrow E_A$: $\neg A_A \lor E_A$→ **C1**
- Fact A_A: A_A→ **C2**
- Negated goal: $\neg E_A$→ **C3**

Clause set: C1: $\neg A_A \lor E_A$C2: A_AC3: $\neg E_A$

4.3 Resolution steps and tree

1. **C1 & C2:** → resolve on A_A: **C4:** E_A
2. **C4 & C3:** → resolve on E_A: **C5:** □

Resolution tree (text):

- □
    - from E_A(C4) and $\neg E_A$(C3)

- E_A from $\neg A_A \lor E_A$(C1) and A_A(C2)

4.4 Conclusion

Empty clause □ ⇒ E_A("Arun is eligible for placement") is proved by resolution.

5. Access Control System

5.1 Propositional representation

Let:

- P**:** "User enters the correct password"
- A**:** "User is authenticated"
- G**:** "User is granted access"

KB:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Goal: Prove G.

5.2 CNF and negated goal

- From $P \rightarrow A$: $\neg P \lor A$→ **C1**
- From $A \rightarrow G$: $\neg A \lor G$→ **C2**
- Fact P: P→ **C3**
- Negated goal: $\neg G$→ **C4**

Clause set: C1: $\neg P \lor A$C2: $\neg A \lor G$C3: PC4: $\neg G$

5.3 Resolution steps

1. **C1 & C3:** → resolve on P: **C5:** A
2. **C2 & C5:** → resolve on A: **C6:** G
3. **C6 & C4:** → resolve on G: **C7:** □

5.4 Conclusion

Empty clause □ ⇒ G("user is granted access") is logically entailed by the KB via resolution.

Summary

Across all five scenarios, the pattern is:

- Represent rules as implications in propositional logic.
- Convert each implication to CNF ($P \rightarrow Q$ becomes $\neg P \lor Q$).
- Add the relevant fact(s) and the **negation of the goal**.

- Use resolution to derive progressively simpler clauses until the **empty clause (□)** appears.

In each case, deriving □ confirms that the original goal is a **logical consequence** of the given knowledge base under the resolution refutation method.